

## IIT JAM 2010 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

| Section / Type        |  | Questions | Marking Scheme | Negative Marking  | Format / Details      |
|-----------------------|--|-----------|----------------|-------------------|-----------------------|
| Multiple Choice (MCQ) |  | 15        | +1 / +2 Marks  | −1/3 / −2/3 Marks | 4 Options (1 Correct) |

## IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at [vaibhavgeometer.github.io](https://vaibhavgeometer.github.io).

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2010 • Q1 • ID: JAM\_2010\_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Which of the following conditions does NOT ensure the convergence of a real sequence  $\{a_n\}$ ?

- (A)  $|a_n - a_{n+1}| \rightarrow 0$  as  $n \rightarrow \infty$
- (B)  $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$  is convergent
- (C)  $\sum_{n=1}^{\infty} n a_n$  is convergent
- (D) The sequences  $\{a_{2n}\}$ ,  $\{a_{2n+1}\}$  and  $\{a_{3n}\}$  are convergent

Question 2 (JAM 2010 • Q2 • ID: JAM\_2010\_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The value of  $\iint_G \frac{\ln(x^2 + y^2)}{x^2 + y^2} dx dy$ , where  $G = \{(x, y) \in \mathbf{R}^2 : 1 \leq x^2 + y^2 \leq e^2\}$ , is

- (A)  $\pi$  (B)  $2\pi$  (C)  $3\pi$  (D)  $4\pi$

Question 3 (JAM 2010 • Q3 • ID: JAM\_2010\_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 The number of elements of  $S_5$  (the symmetric group on 5 letters) which are their own inverses equals

- (A) 10 (B) 11 (C) 25 (D) 26

Question 4 (JAM 2010 • Q4 • ID: JAM\_2010\_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let  $S$  be an infinite subset of  $\mathbf{R}$  such that  $S \cap \mathbf{Q} = \emptyset$ . Which of the following statements is true?

- (A)  $S$  must have a limit point which belongs to  $\mathbf{Q}$
- (B)  $S$  must have a limit point which belongs to  $\mathbf{R} \setminus \mathbf{Q}$
- (C)  $S$  cannot be a closed set in  $\mathbf{R}$
- (D)  $\mathbf{R} \setminus S$  must have a limit point which belongs to  $S$

Question 5 (JAM 2010 • Q5 • ID: JAM\_2010\_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 Let  $f: (1, 4) \rightarrow \mathbf{R}$  be a uniformly continuous function and let  $\{a_n\}$  be a Cauchy sequence in  $(1, 2)$ . Let  $x_n = a_n^2 f(a_n^2)$  and  $y_n = \frac{1}{1+a_n^2} f(a_n^2)$ , for all  $n \in \mathbf{N}$ . Which of the following statements is true?

- (A) Both  $\{x_n\}$  and  $\{y_n\}$  must be Cauchy sequences in  $\mathbf{R}$
- (B)  $\{x_n\}$  must be a Cauchy sequence in  $\mathbf{R}$  but  $\{y_n\}$  need not be a Cauchy sequence in  $\mathbf{R}$
- (C)  $\{y_n\}$  must be a Cauchy sequence in  $\mathbf{R}$  but  $\{x_n\}$  need not be a Cauchy sequence in  $\mathbf{R}$
- (D) Neither  $\{x_n\}$  nor  $\{y_n\}$  needs to be a Cauchy sequence in  $\mathbf{R}$

Question 6 (JAM 2010 • Q6 • ID: JAM\_2010\_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 Let  $\vec{F} = 2xyz e^{x^2} \hat{i} + z e^{x^2} \hat{j} + y e^{x^2} \hat{k}$  be the gradient of a scalar function. The value of  $\int_L \vec{F} \cdot d\vec{r}$  along the oriented path  $L$  from  $(0, 0, 0)$  to  $(1, 0, 2)$  and then to  $(1, 1, 2)$  is

- (A) 0
- (B)  $2e$
- (C)  $e$
- (D)  $e^2$

Question 7 (JAM 2010 • Q7 • ID: JAM\_2010\_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 Let  $\vec{F} = xy \hat{i} + y \hat{j} - yz \hat{k}$  denote the force field on a particle traversing the path  $L$  from  $(0, 0, 0)$  to  $(1, 1, 1)$  along the curve of intersection of the cylinder  $y = x^2$  and the plane  $z = x$ . The work done by  $\vec{F}$  is

- (A) 0
- (B)  $\frac{1}{4}$
- (C)  $\frac{1}{2}$
- (D) 1

Question 8 (JAM 2010 • Q8 • ID: JAM\_2010\_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Let  $\mathbf{R}[X]$  be the ring of real polynomials in the variable  $X$ . The number of ideals in the quotient ring  $\mathbf{R}[X]/(X^2 - 3X + 2)$  is

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Question 9 (JAM 2010 • Q9 • ID: JAM\_2010\_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Consider the differential equation  $\frac{dy}{dx} = ay - by^2$ , where  $a, b > 0$  and  $y(0) = y_0$ . As  $x \rightarrow +\infty$ , the solution  $y(x)$  tends to

(A) 0

(B)  $\frac{a}{b}$ (C)  $\frac{b}{a}$ (D)  $y_0$ 

Question 10 (JAM 2010 • Q10 • ID: JAM\_2010\_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Consider the differential equation  $(x + y + 1)dx + (2x + 2y + 1)dy = 0$ . Which of the following statements is true?

(A) The differential equation is linear

(B) The differential equation is exact

(C)  $e^{x+y}$  is an integrating factor of the differential equation

(D) A suitable substitution transforms the differentiable equation to the variables separable form

Question 11 (JAM 2010 • Q11 • ID: JAM\_2010\_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let  $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$  be a linear transformation such that  $T((1, 2)) = (2, 3)$  and  $T((0, 1)) = (1, 4)$ . Then  $T((5, 6))$  is

(A)  $(6, -1)$ (B)  $(-6, 1)$ (C)  $(-1, 6)$ (D)  $(1, -6)$ 

Question 12 (JAM 2010 • Q12 • ID: JAM\_2010\_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 The number of  $2 \times 2$  matrices over  $\mathbf{Z}_3$  (the field with three elements) with determinant 1 is

(A) 24

(B) 60

(C) 20

(D) 30

Question 13 (JAM 2010 • Q13 • ID: JAM\_2010\_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 The radius of convergence of the power series  $\sum_{n=0}^{\infty} a_n z^{n^2}$ , where  $a_0 = 1$ ,  $a_n = 3^{-n} a_{n-1}$  for  $n \in \mathbf{N}$ , is

- (A) 0 (B)  $\sqrt{3}$  (C) 3 (D)  $\infty$

Question 14 (JAM 2010 • Q14 • ID: JAM\_2010\_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 Let  $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$  be the linear transformation whose matrix with respect to the standard basis  $\{e_1, e_2, e_3\}$  of  $\mathbf{R}^3$  is  $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ . Then  $T$

- (A) maps the subspace spanned by  $e_1$  and  $e_2$  into itself  
 (B) has distinct eigenvalues  
 (C) has eigenvectors that span  $\mathbf{R}^3$   
 (D) has a non-zero null space

Question 15 (JAM 2010 • Q15 • ID: JAM\_2010\_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 Let  $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$  be the linear transformation whose matrix with respect to the standard basis of  $\mathbf{R}^3$  is  $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$ , where  $a, b, c$  are real numbers not all zero. Then  $T$

- (A) is one-to-one  
 (B) is onto  
 (C) does not map any line through the origin onto itself  
 (D) has rank 1

## OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2010 Official Paper • All Official Answers & Acceptance Ranges

| Q. No. | Type | Marks | Official Key |
|--------|------|-------|--------------|
| 1      | MCQ  | +6    | A            |
| 2      | MCQ  | +6    | B            |
| 3      | MCQ  | +6    | D            |
| 4      | MCQ  | +6    | D            |
| 5      | MCQ  | +6    | A            |
| 6      | MCQ  | +6    | B            |
| 7      | MCQ  | +6    | C            |
| 8      | MCQ  | +6    | C            |

| Q. No. | Type | Marks | Official Key |
|--------|------|-------|--------------|
| 9      | MCQ  | +6    | B            |
| 10     | MCQ  | +6    | D            |
| 11     | MCQ  | +6    | A            |
| 12     | MCQ  | +6    | A            |
| 13     | MCQ  | +6    | B            |
| 14     | MCQ  | +6    | C            |
| 15     | MCQ  | +6    | C            |

### SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ( $-1/3$  for 1-mark questions,  $-2/3$  for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g.,  $[a, b]$ ) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.